

Birla Institute of Technology and Science, Pilani
Digital Signal Processing (ECE/EEE F434)

Mid-Semester Examination, First Semester 2025 – 26

Time: 90 minutes

07/10/2025

Maximum Marks: 60

1. You are supposed to decipher the marking scheme for the rest of the questions by solving the following parts of this question, and that will also give you **12** marks.

- Let $x[n] = [1, 2, 2, 1, 2, 1, 1, 2]$. Compute the mod-4 reduction and call it as $x_1[n]$, then evaluate the 4-point DFT of $x_1[n]$ using matrix method. Finally, evaluate the 4-point IDFT of the result. The value of the second sample of IDFT gives

3 (?)

- Let $x_1[n] = [1, 2, 3, 4]$ and $x_2[n] = [1, 0, -1, 1]$, and $x_3[n]$ denotes the circular

4 convolution of $x_1[n]$ and $x_2[n]$. Then the length of $x_3[n]$ gives (??).

- Find the 4-point DFT of a sequence given by $x(n) = 3\delta[n] - 2j\delta[n-1] + \delta[n-8]$ 2] + 2j\delta[n-3]. Then, $X[0]+X[2]$ gives (???)

2. DSP requires mathematical operations, this problem checks your understanding of that. The following claims to be a mathematical proof of $4 = 5$. Which of the steps is/are illegal, or are you convinced that this is correct? Explain in either of the case.

Step 1 : $16 - 36 = 25 - 45$

Step 2 : $4^2 - 9 \times 4 = 5^2 - 9 \times 5$

Step 3 : $4^2 - 9 \times 4 + 81/4 = 5^2 - 9 \times 5 + 81/4$

Step 4 : $(4 - 9/2)^2 = (5 - 9/2)^2$

Step 5 : $4 - 9/2 = 5 - 9/2$

Step 6 : $4 = 5$

(?)

3. Justify/Evaluate the following with proper reasoning, no marks if no justification.

- (a) In DSP, we often say that the phase of a signal has more important perceptual information. Explain, when the magnitude also gives information which is as relevant as the phase information.

(?)

- (b) The Nyquist criterion, regarding the sampling of lowpass signals, is sometimes stated as, "The sampling rate f_s must be equal to or greater than twice the highest spectral component of the continuous signal being sampled." Can you think of how a continuous sinusoidal signal can be sampled in accordance with that Nyquist criterion definition to yield all zero-valued discrete samples?

(?)

- (c) Consider the stream of symbols: $X(0) = 1$, $X(1) = 0$, $X(2) = -1$ and $X(3) = j$, in an OFDM transmitter system, after the modulation. Determine the generated OFDM samples without adding the cyclic prefix?

(??)

- (d) Consider 8-point signal $x[n] = [6, 1, 0, 1, 6, 1, 0, 1]$. Find two other signals, one of length-9 and one of length-10, that have the same 8-point DFT as $x[n]$. These signals must not begin or end with zeros.

(??)

- (e) Consider a sequence with z-transform $X(z) = P(z)/Q(z)$, where $P(z)$ and $Q(z)$ are polynomials in z . If the sequence is absolutely summable and if all the roots of $Q(z)$ are inside the unit circle, then is the sequence necessarily causal? If your answer is yes, clearly explain. If your answer is no, give a counterexample.

(?)

4. Consider an ideal low pass filter with the impulse response $h[n]$ and the frequency response given by

$$H(e^{jw}) = \begin{cases} 1, & |w| < 0.2\pi, \\ 0, & 0.2\pi \leq |w| \leq \pi. \end{cases}$$

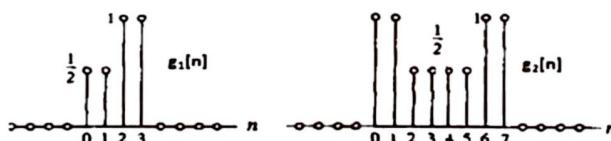
A new filter is defined by the equation $h_1[n] = (-1)^n h[n]$. Determine the frequency response $H_1(e^{jw})$, and plot the same for $|\omega| < \pi$. What kind of filter is this? (???)

5. Consider $x[n] = [1, 1, 1, 1, 1, 1, 1]$, whose DTFT is denoted as $X(e^{jw})$. For the DFT sequence $X_1[k]$ which is obtained by sampling $X(e^{jw})$ at uniform intervals of $\pi/3$ starting from $w = 0$, determine the IDFT $x_1[n]$ of $X_1[k]$ without computing $X(e^{jw})$ and $X_1[k]$. Can you recover $x[n]$ from $x_1[n]$? (???)

6. Claim: Given program performs the circular convolution. Verify if the given claim is correct or incorrect with the necessary justification. (??)

```
function z = circonv(x,y);
N = length(x);
if (length(y) ~= N),
    error('-----');
end
z = conv(reshape(x,1,N),reshape(y,1,N));
z = z(1:N) + [z(N+1:2*N-1),0];
```

7. Let $G_1(e^{jw})$ denote the DTFT of the sequence $g_1[n]$, express the DTFT of a sequence $g_2[n]$ in terms of $G_1(e^{jw})$, no need to evaluate $G_1(e^{jw})$. (??)



8. If you have an FFT subroutine for computing a length- N DFT, the inverse DFT of an N -point sequence $X[k]$ can be implemented using this subroutine as follows:

1. Swap the real and imaginary parts of each DFT coefficient $X[k]$.
2. Apply the FFT routine to this input sequence.
3. Swap the real and imaginary parts of the output sequence.
4. Scale the resulting sequence by $1/N$ to obtain the sequence $x[n]$, corresponding to the inverse DFT of $X[k]$.

Determine whether this procedure works as claimed. If it doesn't, propose a simple modification that will make it work. (??)